Homework

DM10 - Points of a tetrahedron - 50 min.

Consider a non-degenerate tetrahedron $ABCD$. The points I, J and K are the respective midpoints of $[AB]$, $[AC]$ and $[AD]$.

1. Express $\overline{IJ}$ in terms of $\overrightarrow{BC}$ and $\overrightarrow{IK}$ in terms of $\overrightarrow{BD}$.
2. Deduce that the planes (IJK) and (BCD) are parallel.
3. Define the point G by $\overrightarrow{AG} = \frac{1}{4}(\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD})$. Does the point G lie in the plane (IJK) ? Justify.
4. For this question only, assume that, in an orthonormal coordinate system, $A(0;0;0)$, $B(4;0;0)$, $C(0;4;0)$ and $D(0;0;4)$. Calculate the lengths IJ, IK and JK.